

GENE THERAPY

Long-term stability of posttranscriptional genetic silencing of BCL11A using a shmiR vector in sickle cell disease

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KEY POINTS

- Stable, long-term induction of HbF using shmiR LVV in SCD with vector dose-dependent knockdown of BCL11A.
- Highly efficient manufacturing and rapid product release of transduced HSPCs in phase 1 study with no vector related safety issues.

Sickle cell disease (SCD) is characterized by chronic hemolysis, and painful vaso-occlusive episodes (VOE). High levels of fetal hemoglobin (HbF) attenuate the disease phenotype. We used a lentivirus vector (LVV) expressing a short hairpin RNA embedded in a microRNA (shmiR) that targets BCL11A in erythrocytes to induce HbF in a first-in-human study in SCD. The purpose of the study was to assess hematopoietic stem/progenitor cells collection, transduction parameters, safety, HbF induction, and durability. Eleven eligible patients with SCD underwent hematopoietic stem cell (HSC) collection. Plerixafor-mobilized peripheral blood HSCs required for manufacturing were obtained in 1 mobilization cycle for 10 of 11 participants, and 11 of 11 patient products were successfully manufactured with a median time to release of product of 39 days. Ten patients were infused with HSCs transduced with the shmiR vector. Engraftment occurred in all 10 patients. At a median follow-up of 58 months (range, 35-82) after infusion, there were no adverse events attributed to the vector. The transduction efficiency was 93.1%. One

patient demonstrated low engraftment of the transduced cells and had suboptimal HbF induction. In the remaining 9 patients, at 2 years after treatment, the peripheral blood demonstrated 71% F cells with 11.9 pg HbF per F cells, and both parameters remained stable in 9 patients with ≥ 48 months follow-up. All patients who had VOEs before gene therapy demonstrated sustained mitigation of pain events. These data demonstrate excellent manufacturing efficiency, efficacy and safety of targeting BCL11A using a shmiR LVV, and long-term durability of the shmiR vector, leading to a pivotal, multisite, phase 2 trial that is currently underway (NCT05353647). This trial was registered at www.clinicaltrials.gov as NCT03282656.

Introduction

Sickle cell disease (SCD) is a genetic condition caused by a mutation of the adult β -globin locus, which affects the function of hemoglobin (Hb) and contributes significant disease burdens globally. At the molecular level, SCD is caused by a mutation in the adult β -globin gene that alters the sixth amino acid position of the protein from a glutamic acid to a valine

($\beta 6\text{Glu} \rightarrow \text{Val}$), thereby generating a structural change and an alteration in the charge at this position. Sickle Hb has a propensity to polymerize with adjacent Hb molecules in the deoxygenated state, leading to red cell membrane damage, shape change, cell rigidity, and reduced oxygen carrying capacity.^{1,2} The consequent small vessel occlusion and organ ischemia leads to acute and chronic pain, anemia from chronic hemolysis, and a variety of long-term complications.

SCD contributes to significant lifelong morbidity and quality-associated life years lost, even in North America.³ Allogeneic transplantation of hematopoietic stem and progenitor cells (HSPCs) can provide curative therapy, but limited donor availability⁴ and the risk of immunologic complications, such as graft-versus-host disease and graft rejection, remain significant barriers for many patients with SCD.⁵⁻⁷ In 2023, the US Food and Drug Administration (FDA) approved autologous stem cell gene therapy (GT) for SCD using 2 different platforms based on gene editing and gene addition. Exagamglogene autotemcel (exa-cel; Casgevy) uses CRISPR-Cas9 editing of an erythroid-specific enhancer of BCL11A to induce fetal Hb (HbF) production.⁸ Lovotibeglogene autotemcel (Lyfgenia) uses a lentivirus vector (LVV) transgene expression of a sickling-resistant, modified β -globin to effect disease modulation.⁹ Both approaches have shown efficacy, and both have platform-specific challenges, including difficulty in obtaining sufficient numbers of HSPCs for genetic manipulation and loss of cells during the manufacturing process.⁸ This can lead to a need for multiple cycles of HSPC mobilization and collection and subsequent delays, sometimes prolonged, in treating patients.

In an alternative approach, we developed an LVV technology targeting BCL11A mRNA using a short hairpin RNA embedded in a microRNA (shmiR) for posttranscriptional gene editing.¹⁰⁻¹² This approach combines the high efficiency and high target cell preservation of LVV transduction with lineage-specific targeting of BCL11A under transcriptional control of β -globin locus regulatory elements in a third-generation, self-inactivating LVV BCH-BB694. This results in γ -globin reinduction while simultaneously and coordinately reducing sickle Hb with broad HbF expression in red cells. Using this technology, we conducted a first-in-human pilot study to determine the safety, durability, and optimal transduction parameters in preparation for a pivotal, multicenter, phase 2 trial. This pilot trial was used to successfully optimize the manufacturing processes that defined the transduction conditions used in the multisite, phase 2, pivotal trial currently underway ([ClinicalTrials.gov](https://clinicaltrials.gov/ct2/show/study/NCT05353647) identifier: NCT05353647). Data on the phase 2 trial will be reported after the primary end point analysis 2 years after infusion of the last participant in that trial. Here, we report the long-term follow-up results of the 10 infused patients in the pilot study.

Methods

The study design and analyses were described previously¹² and were performed with minor modifications and approval by the institutional review boards of the participating institutions and under an investigational new drug application that was approved by the FDA with safety monitoring by an independent data and safety monitoring board assigned by the National Heart, Lung, and Blood Institute. The data cutoff was 20 October 2025.

A subset of data reported here was initially reported by Esrick et al (supplemental Methods, available on the *Blood* website).¹² Cell manufacturing was performed as previously described.¹² The vector multiplicity of infection (MOI) and the use of transduction enhancers (TEs) were modified during the trial to obtain an appropriate in vivo vector copy number (VCN;

supplemental Table 3). The final formulation used for patient BCL-012 included only Poloxamer 338 at 1 mg/mL and an MOI of 0.6.

The full clinical protocol is available.¹² The protocol was approved by the requisite regulatory authorities and registered at [ClinicalTrials.gov](https://clinicaltrials.gov/ct2/show/study/NCT03282656) (identifier: NCT03282656). All authors had access to the primary data. Cell transduction was performed at 2 sites, including at the Dana-Farber Cell Manufacturing Core Facility, and as described¹³ using good manufacturing practice conditions. Additional methods are described in the supplemental Materials.

This study was conducted in accordance with the Declaration of Helsinki.

Results

We conducted a first-in-human, phase 1, dose-finding, pilot, and feasibility study (NCT03282656) testing the safety of GT using an LVV expressing a shmiR targeting BCL11A to induce HbF expression in patients with SCD. Our protocol treatment schema is shown in supplemental Figure 1. This was an open-label, multisite, single-arm, cohort study of a single infusion of autologous bone marrow-derived CD34⁺ hematopoietic stem cell (HSC) cells transduced with the LVV containing a short hairpin RNA targeting BCL11A. The primary end point was neutrophil engraftment within 7 weeks after infusion.

Eleven eligible participants with SCD (HbSS or HbS/Hb β ⁰, including 1 patient with 2-gene deletion α -thalassemia) underwent HSPC collection. One patient withdrew before infusion because of progression of comorbidities during the time between consenting and product release, and 10 were infused with autologous HSPCs transduced with the shmiR vector. The baseline characteristics of the eligible participants are shown in supplemental Table 1. Of the 11 enrolled patients, 5 were receiving chronic transfusions (3 for a history of overt stroke and 2 for silent stroke and pain), and the other 6 were on hydroxyurea before enrollment. The number of severe vaso-occlusive episodes (VOEs; defined as sickle cell pain requiring admission for opioid medications, priapism requiring admission, or acute chest syndrome requiring admission) in the 24 months before enrollment among the 5 patients who did not receive chronic transfusions ranged from 3 to 7. One patient had a history of recurrent priapism requiring frequent hospitalizations.

Peripheral blood (PB) mobilization was accomplished with plerixafor administration, as previously described,¹⁴ and HSPCs were collected by apheresis. Plerixafor was dosed daily at 240 μ g/kg subcutaneously between 3 and 4 hours before the onset of collection with the goal of collecting at least 5 blood volumes or having up to 8 hours on the apheresis instrument. CD34 HSPCs were isolated using a CliniMACS device, and the cells were transduced following good manufacturing practices, as previously described.¹³ Because this was a dose-finding pilot study, changes in MOI and the number of TEs were adjusted based on the VCN of each previous product and infused participant. Apheresis collection outcomes and final manufactured cell doses are shown in Figure 1. Mobilized PB

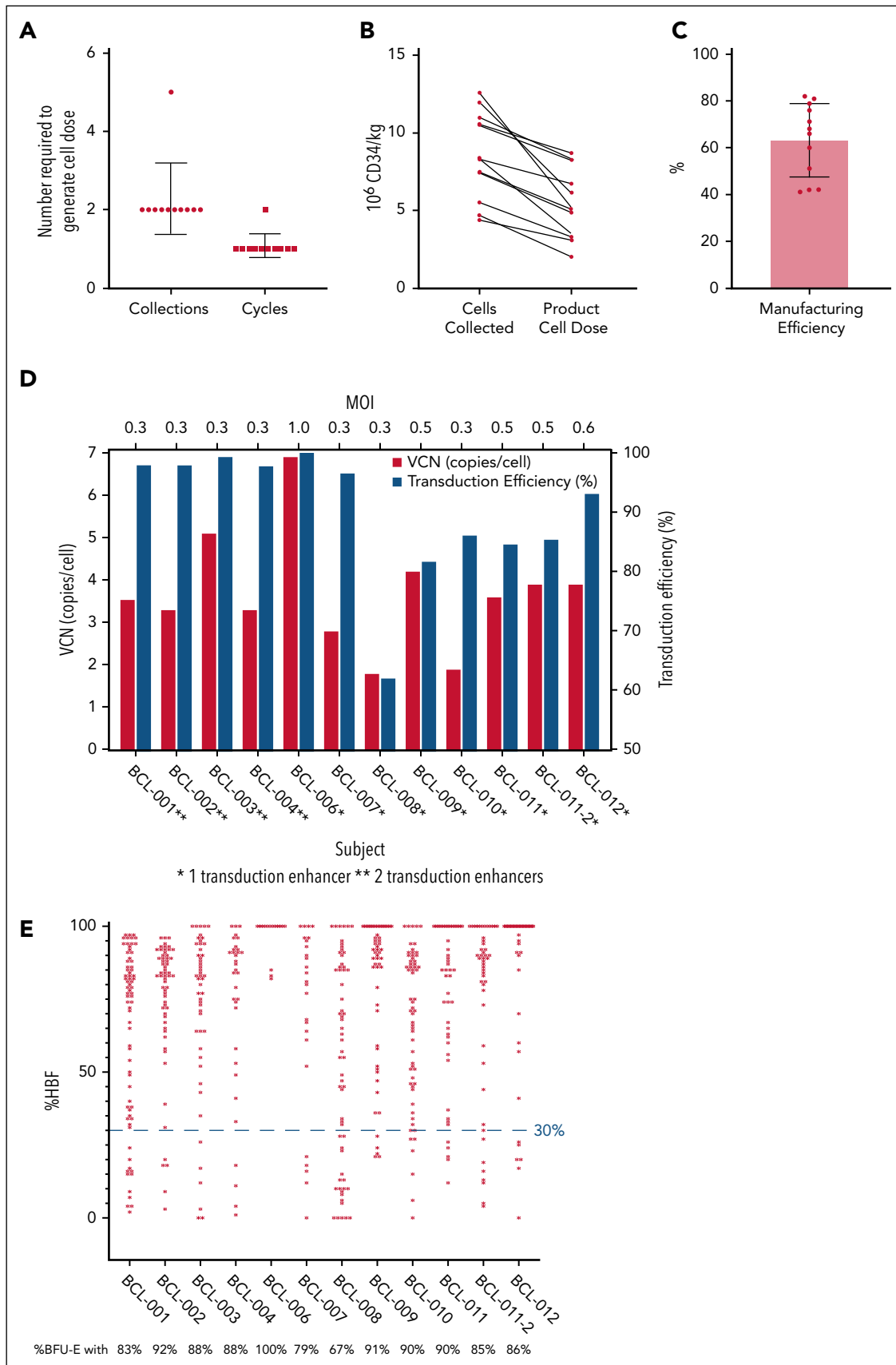


Figure 1. Cell processing and manufacturing efficiency. (A) Mean number of collections (individual apheresis procedures) and collection cycles (2 apheresis procedures on consecutive days) required to achieve the target cell dose. Each dot represents an individual participant. (B) Reduction of CD34 HSPCs (10^6 /kg) during purification and transduction process. (C) Manufacturing efficiency (calculated as total product cell dose divided by total cells collected by apheresis). In B and C, each dot represents an

HSPCs required for manufacturing were obtained in 1 mobilization cycle (2 consecutive days of apheresis collection) for 10 of 11 participants with 1 participant requiring 2 cycles. A median of 1 (mean of 1.1) mobilization cycle overall and a median of 2 collection days per participant (mean of 2.3) were required for product manufacturing (Figure 1A; supplemental Table 2). The number of mobilization cycles, days of apheresis collection, CD34 HSPCs collected for manufacturing, CD34 HSPCs in the final product, and postmanufacturing percentage recovery are shown in Figure 1B and for individual participants in supplemental Table 2. A backup, unmanipulated product of $\geq 2 \times 10^6$ CD34 cells per kg was generally obtained by apheresis at the end of the collection admission but, in some cases, at the start of admission for product infusion. On average, there was a loss of 3.1×10^6 CD34 cells per kg during purification and transduction (Figure 1B). The mean manufacturing efficiency, defined as the number of CD34 HSPCs in the final product divided by the number of CD34 HSPCs collected, was 63% (range, 41%-82%; Figure 1C). The median overall final cell product count was 5.5×10^6 (range, 3.3×10^6 to 8.7×10^6 ; N = 11) CD34 HSPC cells per kg with 1 product (BCL-009) infused after an FDA-approved planned deviation due to the lower cell number. The median time from the first day of apheresis to final release of the product was 39 days (range, 22-151; N = 11, including 1 participant with 2 manufactured products) with only 1 product released at >50 days after apheresis. To define optimal transduction conditions, MOI and the use of TEs varied during the trial (Figure 1D; supplemental Table 3), with the last 7 participants' transductions using 1 TE. Overall, the median transduction efficiency for all products based on p24⁺ progenitor colonies was 93.1% (range, 62%-100%; N = 11), with a VCN of the product of 3.7 ± 1.4 (mean $\pm$ standard deviation [SD]; median, 3.6 [range, 1.8-6.9]; N = 11; Figure 1D). This final transduction procedure was selected for use in the subsequent phase 2 trial (GRASP, NCT 05353647), which included an MOI of 0.6 and only 1 TE, Poloxamer 338, at 1 mg/mL with the VCN and transduction efficiency shown in Figure 1D for patient BCL-012.

As part of the investigational new drug (IND)-defined product release criteria, we analyzed vector-induced HbF expression in erythroid hematopoietic progenitor colonies derived and cultured from the manufactured cell product (Figure 1E). BCL11A was effectively targeted leading to an induction of HbF with a median of 88% (range, 67%-100%) of individual burst-forming unit-erythroid colonies of the cell product containing $\geq 30\%$ of HbF, as determined by high-performance liquid chromatography analysis.

After myeloablative busulfan conditioning and product infusion, all patients achieved prompt engraftments (neutrophils: median, 22 days [range, 18-30]; platelets: median, 34 days [range, 26-62]; supplemental Table 4). Vector insertion site analysis (ISA) was monitored in all participants. No single ISA-defined clone had an abundance of > 3% for any participant at any time during follow-up (supplemental Figure 2). The

estimated population size of engrafted transduced HSCs over time was stable after 6 months based on ISA and the use of Chao1 to estimate minimum population sizes (supplemental Figure 3). After initiation of the trial and following the report of leukemia and myelodysplastic syndrome in 2 group A participants in the bluebird bio Lentiglobin HGB-206 gene addition trial for SCD,^{15,16} clonal hematopoiesis analysis for each participant was performed at baseline and over time on PB and bone marrow. Because of the initiation of this analysis after the trial had already begun, previously obtained and banked samples were also evaluated as part of this screening for some participants. A clinical laboratory improvements amendment (CLIA)-validated panel of 88 genes (termed the Rapid Heme Panel) was used for screening.¹⁷ As shown in supplemental Table 5, there were presumed germ line variations of unknown significance detected in genes *SETD2*, *ASXL1*, and *CREBBP*, but no new somatic mutational clones emerged in any participant after GT. One participant (BCL-011) had a pathogenic *DNMT3A* mutation noted 6 months after infusion that was retrospectively determined to have been present before cell manufacturing but not initially reported because it was below validated detection limits of the assay in the clinical laboratory. The variant allele fraction of this mutation increased after infusion to a maximum of 9.9% at 4.5 years of follow-up with no evidence of dysplasia in the 2-year post-GT bone marrow. Subsequent research studies by our group demonstrated extremely low levels of clonal hematopoiesis mutations before and after GT manipulations in 6 patients studied in detail from the pilot study with no consistent effect from either vector insertion or cell manipulation and engraftment.¹⁸ These mutations were well below allele fractions validated for clinical use. Other studies have provided contradictory evidence of underlying clonal hematopoiesis mutations in patients with SCD.^{19,20}

All adverse events (AEs) $\geq$ grade 2 until month 12 after GT and only those considered attributable to GT treatment after month 12 were recorded. There were no AEs attributable to the GT product. Most AEs observed were consistent with the known side effects of obtaining vascular access, apheresis, collection, busulfan conditioning, and the transplant process. Supplemental Table 6 shows a summary of the recorded grade 3 and 4 AEs. Grade 3 events occurring at >10% frequency included mucositis, alopecia, transaminitis, anorexia, and electrolyte disturbances. Grade 4 events were limited to cytopenias following conditioning. Severe AEs after discharge from transplant and up to 2 years after GT comprised 3 <24-hour admissions to rule out bacterial infection (3 patients), 2 admissions for priapism recurrence (1 patient), 1 admission for weight loss and evaluation of gastrointestinal illness (1 patient), and 11 admissions for pain, including 3 with concomitant acute chest syndrome (4 patients). All study severe AEs are listed in supplemental Table 9.

The PB hematologic values for each participant at 6, 12, and 24 months and at last follow-up after infusion are shown in

Figure 1 (continued) individual collection. (D) VCN in copies per cell (red bars) and transduction efficiency (percent) of the drug product (blue bars) by patient. Participants are listed along the x-axis with the number of TE denoted (*1TE, **2 TE). (E) Percentage HbF in individual burst-forming unit-erythroid (BFU-E) in drug products. Each dot represents the percentage HbF measured in a single BFU-E colony derived from the indicated drug product samples (n = 12 cycles from n = 11 eligible participants). The dashed blue line marks the 30% HbF threshold. Numbers below each column indicate the percentage of BFU-E colonies with $\geq 30\%$ HbF.

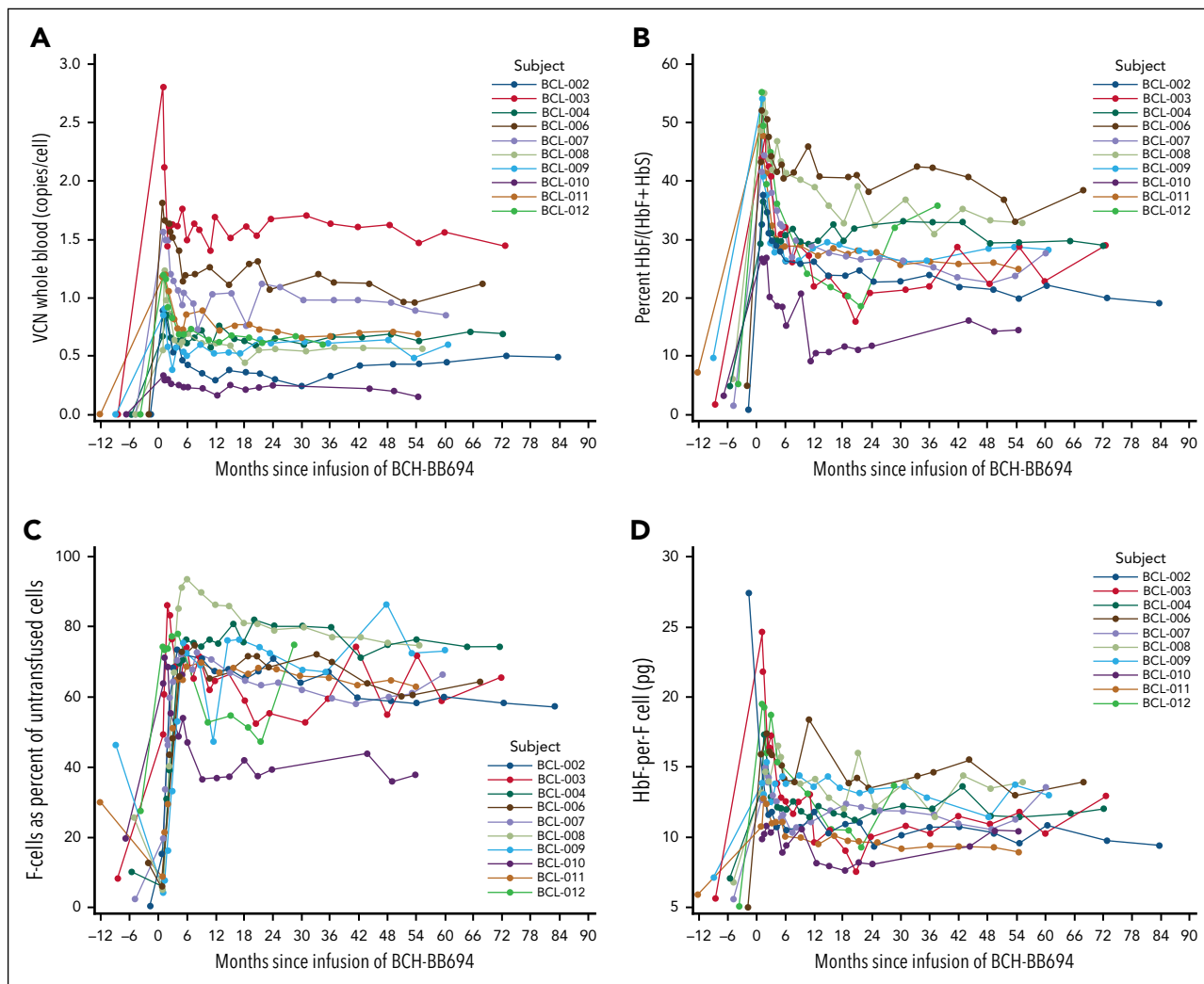


Figure 2. Induction of HbF for each participant over time. (A) VCN in whole blood over time for each patient (copies per cell). (B) PB HbF over time expressed as percentage of F/HbF + HbS. (C) Percentage of F cells over time (calculated as the percentage of untransfused cells). (D) HbF per F cells (in picograms of Hb). Each individual patient is shown from before infusion until last follow-up with infusion of gene-modified cells at time point 0 on the x-axis. At 54 months postinfusion BCL-007 was started on hydroxurea by his local hematologist.

supplemental Table 7. Eight participants demonstrated an increase in baseline Hb, 7 out of 10 participants had a Hb at most recent check of either > 11 g/dL or an increase of > 3 g/dL compared to baseline, and there were reductions in the lactate dehydrogenase levels and reticulocyte counts. Although improved, a lower level of hemolysis continued in all participants. BCL-011, who had 2 α -globin gene deletions, had an ~2 g/dL increase in Hb after infusion, and the participant with the lowest in vivo VCN (BCL-010) had the smallest incremental increase in Hb after GT to ~8 g/dL.

The VCN was stable in PB over the entire follow-up period with a median of 0.7 copies per diploid genome (c/dg; range, 0.2–1.5) in whole blood starting at 6 months after infusion (Figure 2A). The VCN in bone marrow (BM) CD34 HSPCs was assessed at 6 months and 2 years after infusion and was stable at the 2-year time point (supplemental Figure 4A). Because BCL11A is an essential transcription factor in B-cell development, we specifically examined VCN stability in CD19 cells (supplemental Figure 4B). Aligned with the vector design to

restrict expression to erythroid cells and avoid expression of the shmiR in B-cell lineages, it remained stable in the long term. The CD19 B-cell counts and immunoglobulin G levels were similar between baseline and 2 years after infusion in all participants (supplemental Table 8). The CD4 and CD8 T-cell numbers were also similar before GT and at 2 years after infusion (supplemental Table 8).

The participants with the lowest in vivo VCN (BCL-010) had suboptimal HbF induction after infusion of 14.1% at 6 months, with levels remaining stable but low over the entire follow-up period (Figure 2A–B). This approximates the lower gene dosage in vivo for effective HbF induction for this vector. Of the remaining 9 infused participants, there was a higher HbF induction at a mean whole blood HbF of 29.0% (range, 20.8%–38.2%), a mean of 70% (range, 55.3%–80.3%) for HbF-containing red cells (F cells; Figure 2C), and a mean HbF per F cells of 11.7 pg (range, 9.4–13.6; Figure 2D) at 2 years that remained stable at 48 months (n = 8). With the modification in manufacturing for the subsequent phase 2 GRASP trial, HbF

Table 1. Transfusions and major clinical events before and after GT

Participant no.	Time from infusion to date of last contact (mo)	No. of transfusions (annualized)*		No. of severe VOEs			
		Prestudy	After engraftment	Prestudy	After GT (0-6 mo)	After GT (6-24 mo)	After GT (>24 mo)
Participants on chronic transfusions before enrollment							
BCL-002	84	12.5	0	0	0	0	1
BCL-003	77	10.0	6.3	0	0	0	0
BCL-007	60	12.0	0	0	0	0	0
BCL-010	61	11.0	0.99	0	0	3	4
BCL-012	38	13.5	6.1	0	0	2	0
Participants not on chronic transfusions before enrollment							
BCL-004	72	2.0	0	5	2	0	1
BCL-006	71	3.0	0	3	0	0	1
BCL-008	55	2.0	0	3	0	0	0
BCL-009	61	3.5	0.6	5	0	1	1
BCL-011	54	0.5	0	7	0	3	1

The data are presented for 10 infused participants.
*Prestudy events were observed during the 2 years preceding study enrollment. Events that occurred after GT were observed from the time from engraftment to the most recent follow-up.

induction appears uniformly better, with a PB HbF mean $\pm$ SD of $44.9\% \pm 8.3\%$ and a PB F-cell content mean $\pm$ SD of $87.9\% \pm 11.2\%$ at 6 months in the initial 20 participants who were infused to date (data not shown).

BCL11A mRNA knockdown compared with each participant's baseline values measured in isolated CD71⁺ CD235a⁺ erythroid progenitors at 6 months and 2 years, was stable at 42% to 74%, except for participant BCL-010, who demonstrated <20% knockdown (supplemental Figure 5A). There was no significant change in BCL11A expression in CD19⁺ B cells (supplemental Figure 5A). At both 6 months and 2 years, the level of BCL11A knockdown was correlated with the VCN at each time point (supplemental Figure 5B).

In this phase 1 pilot and feasibility study, we gathered efficacy data, in addition to toxicity data, despite the heterogeneity of the treated participants' phenotypes upon entering this trial. A total of 20 severe VOEs occurred in 10 participants infused with GT (Table 1). A post hoc analysis divided the 10 infused participants into those on chronic red blood cell transfusions before GT (n = 5; see supplemental Table 1 for indications for chronic transfusions) and those who were not (n = 5). For 5 participants who were not chronically transfused before GT, the annualized number of VOEs (mean $\pm$ SD) decreased from 2.3 ± 0.8 during the 24 months before GT to 0.5 ± 0.9 during the 6 to 24 months after GT and to 0.2 ± 0.1 for the >24 months after GT. Ten severe VOEs occurred within these 5 participants who were not on chronic transfusions before enrollment: 2 VOEs were reported at <6 months after GT, 4 at 6 to 24 months after GT, and 4 at >24 months after GT (Table 1).

For 5 participants on chronic transfusions before GT, the annualized number of red cell transfusions (mean $\pm$ SD) in the 24 months before GT was 11.8 ± 1.4 , which decreased to 2.7 ± 3.2 over a median post-GT follow-up of 61 months (range, 38-84). One participant (BCL-003), as planned before enrollment, was maintained on red cell transfusions due to preexisting moyamoya with the goal of keeping HbS <40% by a combination of increased HbF and transfusions with HbA of ~60%. Compared with the pre-GT period, this participant's transfusion requirement was reduced by ~50%. Ten severe VOEs occurred within 5 patients on chronic transfusions before enrollment, with 5 events occurring at 6 to 24 months after GT and 5 occurring at >24 months after GT. Seven of these occurred in 1 participant (BCL-010) with the lowest in vivo VCN and HbF induction.

Initially, a history of prior stroke risk was not an exclusion criterion in this study; after initiation of the trial, patients with severe structural neurovascular disease were excluded. Participants' baseline history of central nervous system events and risk factors were variable and included participants with overt stroke, silent stroke, or neither before enrollment (Table 2). Available pre-GT and post-GT stroke history and neuroimaging data are shown in Table 2. All participants had magnetic resonance imaging (MRI) and magnetic resonance angiography (MRA) performed as pretransplant standard of care. Post-GT neuroimaging was not directed by the study protocol, but a portion of the patients had MRI studies performed as part of clinical care. All MRIs from immediately before GT and all post-GT brain images were reviewed by 1 experienced senior neuroradiologist. None of the 10 treated participants had experienced an overt stroke since GT. Three of the 10 treated

Table 2. Major CNS clinical outcomes by participant

Participant	Age at GT (y)	Time from GT (y)	Stroke risk history			Transfusion regimen before GT			After GT			
			Abnormal TCDs	Silent strokes	Overt strokes (age)	Y/N	Time before GT (y)	Indication	Transfusions after GT	Overt stroke after GT	Brain MRI (years after GT for most recent MRI)	New silent strokes after GT
BCL-002	21	7	n/a	N	Y (3 y)	Y	17	History of overt stroke	N	N	Y (6)	N
BCL-003*	26	6	n/a	N	Y (2 y, 16 y)	Y	24	History of overt stroke	Y*	N	Y (6)	N
BCL-004	24	6	N	N	N	N	n/a	n/a	N	N	N	n/a
BCL-006	16	6	N	N	N	N	n/a	n/a	N	N	N	n/a
BCL-007	12	5	N	N	Y (1 y)	Y	11	History of overt stroke	N	N	Y (5)	N
BCL-008	7	4	N	N	N	N	n/a	n/a	N	N	N	n/a
BCL-009	13	5	N	N	N	N	n/a	n/a	N	N	Yes (1)	N
BCL-010	20	5	N	Y	N	Y	3	Silent strokes noted	N	N	Y (3)	N
BCL-011	23	4	N	Y	N	N	n/a	n/a	N	N	Y (5)	N
BCL-012	23	3	N	Y	N	Y	6	Previous silent strokes, pain	Y†	N	Y (1)	N

The data are reported for 10 infused participants.
n/a, not applicable; N, no; Y, yes.
*Patient 3: 1994 overt stroke and started transfusion regimen; 2009 (while off transfusions on research study of hydroxyurea for stroke prevention) had another stroke; 2010 pial synangiosis for moyamoya. Transfusions continued after GT because of high stroke risk.
†Patient 12 was on transfusions after GT from ~6 to 26 months due to hepatopathy event, not related to stroke risk.

participants had a history of overt stroke in early childhood and had been on chronic transfusions for >10 years (range, 11-24) before GT. One of those 3 participants with overt stroke (BCL-003) had severe vasculopathy and continued transfusions, as described previously, and post-GT imaging in this patient, most recently MRI and MRA almost 7 years after GT, has remained stable. Two of the 3 participants with overt stroke had no vasculopathy and have received no transfusions since GT. Post-GT MRIs of these 3 participants with overt stroke showed no new silent cerebral infarcts (SCI) at the most recent follow-up imaging at 4 to 6 years after GT. Three of the other 10 treated participants had evidence of SCI on pre-GT MRIs. The post-GT MRIs in 2 of these 3 participants with SCI (at 3-5 years after GT) showed no progression of SCI. The other participants with SCI had not yet had an MRI off transfusions after GT. Four of the 10 participants who were treated had neither a history of elevated transcranial doppler (TCD) screening, overt stroke, nor SCI on pre-GT MRI. One of these 4 had a post-GT clinical MRI with no finding of SCI.

Discussion

GT for SCD is transformative, and there are currently multiple approaches designed to correct the pathologies arising from the underlying genetic aberration. There is a lack of robust data regarding the differences in efficacy and toxicity between the different approaches and it may require years of follow-up to understand.²¹ The assessment of efficiency, however, is more readily measurable. In this long-term follow-up of a first-in-human phase 1 dose-finding study, we report that the LVV shmiR platform is associated with expedited and efficient manufacturing, thereby enabling patients to receive GT by 6 months after initial enrollment or earlier if the patient is on a chronic transfusion regimen.

A clear advantage of the LVV approach is that fewer cells need to be collected for manufacturing, thereby reducing the number of mobilization and collection cycles and significantly reducing the time from collection to infusion of the GT product. Most patients (9/10) in this study required only 1 collection cycle with a mean manufacturing recovery of >60% of cells collected and a final cell product of $>5 \times 10^6$ CD34 cells per kg, demonstrating less toxicity to HSCs with the use LVV transduction compared with electroporation currently required for gene editing approaches. Published data on gene editing leading to FDA approval of exa-cel described a median of 2 collections (range, 1-6) required to obtain a drug product.⁸ Because each cycle requires an ~4-week recovery period, patients may require up to 6 months to obtain adequate cell numbers for manufacturing for gene editing and additional months for manufacturing and release, which can lead to a waiting period of up to a year between initiation of pre-collection red cell transfusion and infusion of the gene-modified cells.²² In this study, the median time from the first day collection to release of the manufactured product was 39 days with only 1 product released >50 days from initial collection. In this regard, it will be important to compare results of the approach described when multiple sites are used with other GT approaches and with evolving real-world data being generated with these approaches.

We have demonstrated that the shmiR vector leads to effective, dose-dependent knockdown of the target BCL11A mRNA

and a concomitant induction of HbF. Knockdown of BCL11A is lineage restricted with stable induction of pancellular HbF over a follow-up >5 years in some participants with no apparent toxicities, including specific assessment of the B-cell lineage.^{10,11} The stability of VCN over the time of follow-up demonstrates no deleterious effects on the HSC compartment. The lack of toxicity in these lineages is noteworthy because BCL11A is an essential transcription factor in B-cell development, and we and others have demonstrated that the expression of BCL11A is critical for HSC function. Post-transcriptional gene editing by mRNA knockdown leads to some residual BCL11A expression and continued HbS expression that may explain ongoing hemolysis. Our data suggest that a minimum in vivo VCN appears to be around 0.2 c/dg, and effective induction of HbF appears to be obtained at VCNs of between 0.5 and 1 with this vector.

Effective transduction and function of the shmiR vector was demonstrated by analyzing individual burst-forming unit-erythroid from the drug production by high-performance liquid chromatography, which demonstrated uniform induction of HbF to >30% of total Hb in these primitive erythroid colonies. This level of HbF induction has been postulated to be required to effectively inhibit HbS polymerization in the deoxygenated environment.^{1,2,23,24} Induction of HbF varied among participants and manufacturing conditions and was clearly suboptimal in 1 participant with an in vivo VCN of 0.2 c/dg. Although this study was designed to examine toxicities of the shmiR vector, we did see significant mitigation of the SCD phenotype. Although demonstrating an increase in Hb, the 1 participant with 2 α -globin gene deletions did not attain an Hb comparable with other participants. Notably, the incomplete response of 1 participant with priapism as a major VOE event. Whether this represents permanent or slowly resolving vascular damage in the penile corpus blood vessels remains to be seen but it does suggest that such VOEs may not respond as well or as quickly as other VOE complications in some patients. Indeed, there are scattered reports that priapism episodes can recur after successful allo-transplantation.^{25,26}

Neurovascular injury remains a major burden for patients with SCD. By the age of 30 years, >50% of adults with sickle cell anemia have SCIs.²⁷ Allogeneic transplants have improved cerebral oxygen parameters, and the risk of progressive cerebrovascular injury visible on MRI is lower after transplant than after chronic transfusions.^{28,29} The role of autologous GT in mitigating central nervous system damage and decreasing the risk for future stroke will need to be examined in a long-term study with robust longitudinal imaging and neurocognitive outcome analysis, which were not included in this phase 1 study. Although only a portion of patients in this study have had post-GT neuroimaging, no patient had an overt stroke, and we have not identified any patients with new SCI. In the ongoing phase 2 study, more detailed analysis of central nervous system outcomes is underway to understand whether GT, rather than chronic red cell transfusions or hydroxyurea, will provide neuroprotection in select patients with an increased stroke risk. Additional follow-up is needed to ensure the adequacy of Hb F induction to protect against stroke evolution.

One advantage of the shmiR vector that targets BCL11A is that, by using a binary physiologic switch, an increase in HbF is accompanied by a concomitant decrease in HbS. This is

important for 2 reasons. The HbS concentration largely drives polymerization in the deoxygenated state,^{23,30,31} which may explain the superior protection from polymerization at low oxygen concentrations seen in patients who received GT compared with patients on hydroxyurea, as we previously reported.²⁴ In addition, compared with gene addition approaches, such as those used for lovotibeglogene autotemcel GT, the coordinated reduction in HbS may mitigate any tendency toward vector-induced imbalance of α -globin and β -globin chains. This might be particularly important for patients with SCD with α -gene deletions, a coinherited condition in some patients. Overall, the small number of participants treated in our pilot study, combined with still relatively short follow-up of some participants, makes definitive comparisons of the safety of these approaches premature. Research studies by our group demonstrated extremely low levels of clonal hematopoiesis mutations before and after GT manipulations in 6 patients who were studied in detail from this pilot study with no consistent effect from either vector insertion or cell manipulation and engraftment.¹⁸ These mutations were well below the allele fractions validated for clinical use. Other studies have provided contradictory evidence of underlying clonal hematopoiesis mutations in patients with SCD.^{19,20} The expansion of a *DMT3A* pathologic mutant in BCL-011 during/after engraftment is consistent with reports of expanded clones of mutants of *DMT3A* after allotransplantation without genetic manipulation of donor cells.

The data generated in this phase 1 study informed manufacturing optimization for the subsequent multicenter phase 2 GRASP study, which has led to uniformly higher and more broadly distributed HbF induction at levels comparable with those achieved in clinical studies of exa-cel. The data of this pilot study should be interpreted with some caution due to the small number of participants who were treated at 1 center. In addition, the different manufacturing conditions, including different MOIs used in the pilot study and the resulting variable in vivo VCNs, further complicates interpretation of the efficacy of the approach. In the initial analysis of the participants in the multisite phase 2 GRASP study, those who reached the 6 months postinfusion mark had an average of 90% F cells and HbF blood levels of >40%, thus strengthening the interpretation of the pilot study. The changes in manufacturing during the phase 1 study included adjusting the MOI to effect efficient transduction of cells in the drug product, and elimination of one of two transduction enhancers that have recently been used in several LVV clinical trials. The optimal MOI and use of only one transduction enhancer have led to simplified manufacturing with higher cell yields and shorter turn-around to product release while keeping the cell collection cycles less ≤ 2 . The multicenter GRASP study has completed enrollment, and follow-up analysis is underway. In conclusion, long-term data from this first-in-human study show exceptional efficiency in the collection and manufacturing of product, as well as durability, safety, and efficacy in mitigating the complications of SCD using the shmiR vector targeting posttranscriptional gene silencing of BCL11A.

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Footnotes

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Deidentified individual participant data are available for an indefinite period from the corresponding author, David A. Williams (DAWilliams@childrens.harvard.edu).

The online version of this article contains a data supplement.

There is a [Blood Commentary](#) on this article in this issue.

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